

$$d1 \quad \frac{X_0}{X_0} \approx \frac{1}{s^2 + s + 1} \quad \because \frac{+2.01}{+1.03} \quad |s| > | \pm 0.5 |$$

(= 所需振荡器)

$$\frac{X_0}{X_0} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$\therefore \begin{cases} \omega_n = 1 \\ \zeta = \frac{1}{2} \end{cases} \quad \therefore \omega_d = \omega_n \sqrt{1 - \zeta^2} = \frac{\sqrt{3}}{2}$$

$$t_p = \frac{\pi}{\omega_d} = \frac{\sqrt{3}}{2} \pi$$

$$t_s = \frac{3}{\zeta\omega_n} = 6$$

$$\textcircled{\text{解}} \quad M_p = e^{-\frac{\zeta\pi}{\sqrt{1-\zeta^2}}} = e^{-\frac{\frac{1}{2}\pi}{\frac{\sqrt{3}}{2}}} = e^{-\frac{\pi}{\sqrt{3}}}$$

$$t_r = \frac{1}{\omega_d} (\pi - \arccos \zeta)$$

$$= \frac{2}{\sqrt{3}} (\pi - \arccos \frac{1}{2})$$

$$\textcircled{Q2} \quad G_{open} = \frac{0.04}{s(s+1)}$$

$$\textcircled{1} \quad G_{closed} = \frac{\frac{0.04}{s(s+1)}}{1 + \frac{0.04}{s(s+1)}}$$

$$= \frac{0.04}{s^2 + s + 0.04} \quad \#$$

$$\text{E(s)} = \lim_{s \rightarrow 0} s E(s)$$

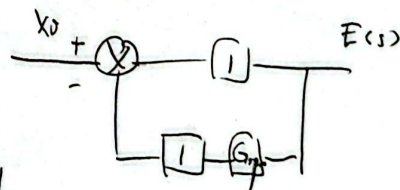
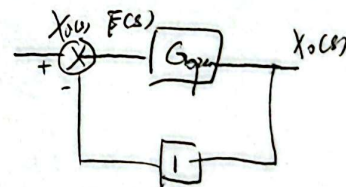
$$\therefore x_0(t) = 1(t), \quad X_0(s) = \frac{1}{s}$$

$$\frac{E(s)}{X_0(s)} = \frac{\frac{1}{1 + G_{open}}}{\frac{1}{s}}$$

②

$$\therefore e_{ss} = \lim_{s \rightarrow 0} s \frac{1}{1 + \frac{0.04}{s(s+1)}} = \lim_{s \rightarrow 0} \frac{s}{s^2 + s + 0.04} = 0 \quad \#$$

$$\textcircled{3} \quad I = \int_0^\infty |a(t)| dt$$



Q3 $G(s) = \frac{as+1}{s^2}$

$$\therefore |G(j\omega)| = \left| \frac{j a \omega + 1}{- \omega^2} \right|$$

$$= \frac{\sqrt{a^2 \omega^2 + 1}}{\omega^2}$$

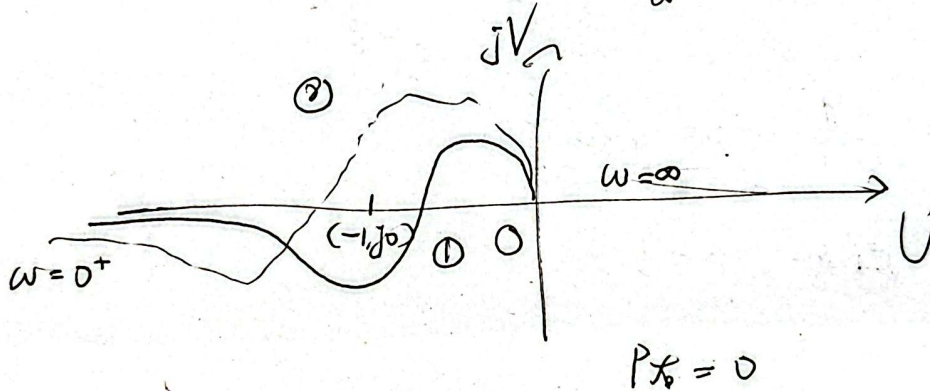
$$\angle G(j\omega) = -180^\circ + \arctan\left(\frac{a\omega}{1}\right)$$

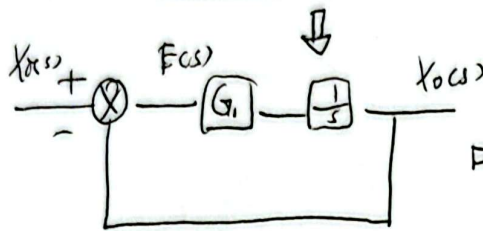
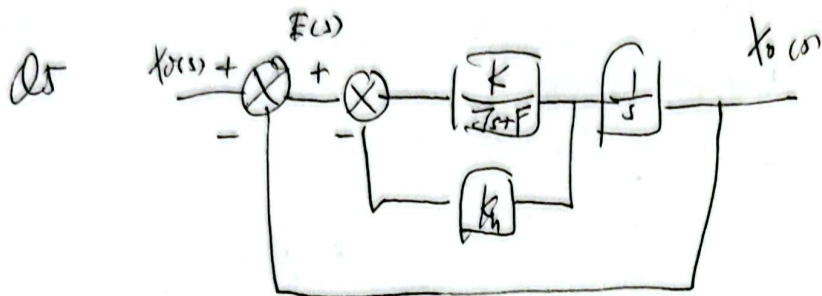
$$\therefore \gamma_0 = 180^\circ + \angle G(j\omega) = \arctan(a\omega)$$

$\neq \gamma_0 = 45^\circ$, ~~$\arctan(a\omega)$~~

~~$\frac{1}{\omega}$~~ $a\omega = 1$
 $\therefore a = \frac{1}{\omega}$

Q4





$$G_1 = \frac{\frac{K}{s+F}}{1 + \frac{K K_h}{s+F}}$$

$$= \frac{K}{s+F+K K_h}$$

$$\Rightarrow \text{Reduced} = \frac{\frac{1}{s} \left(\frac{K}{s+F+K K_h} \right)}{1 + \frac{1}{s} \left(\frac{K}{s+F+K K_h} \right)}$$

$$= \frac{K}{s^2 + (F+K K_h)s + K}$$

① 稳定条件: $\zeta > 0$

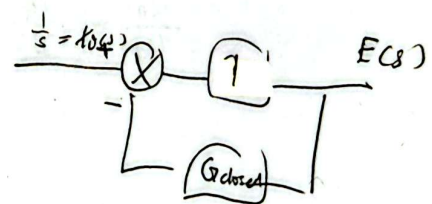
$$\begin{cases} (F+K K_h)s = (2\zeta\omega_n)s \\ K = \omega_n^2, \therefore \omega_n = \sqrt{K} \end{cases}$$

$$\therefore \frac{F+\sqrt{K} K_h}{2\sqrt{K}} = \zeta > 0$$

② ~~X_o(t) = 1(t)~~ $X_o(t) = 1(t)$, $X_o(s) = \frac{1}{s}$

$$\therefore \frac{E(s)}{X_o(s)} = \frac{\frac{1}{1+G_{closed}(s)}}{\frac{1}{s}}$$

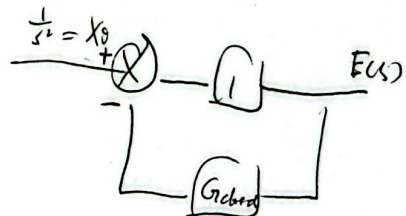
$$e_{ss1} = \lim_{s \rightarrow 0} s \left(\frac{1}{1+G_{closed}} \right)$$



③ $X_o(t) = t^n(t)$, $X_o(s) = \frac{1}{s^{n+1}}$

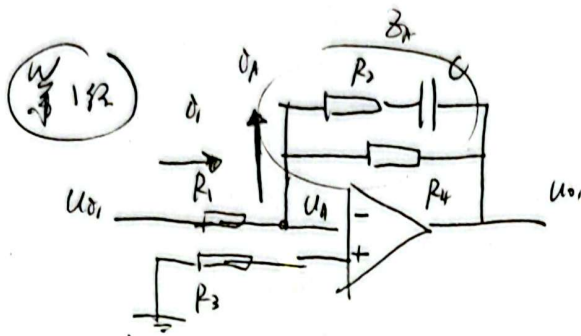
$$\frac{E(s)}{X_o(s)} = \frac{\frac{1}{1+G_{closed}(s)}}{\frac{1}{s^{n+1}}}$$

$$e_{ss2} = \lim_{s \rightarrow 0} s^n \left(\frac{1}{1+G_{closed}} \right)$$



$$\frac{t^n}{s^{n+1}} \Rightarrow \text{Laplace}[t^n \cdot 1(t)] = \frac{n!}{s^{n+1}}$$

Q6



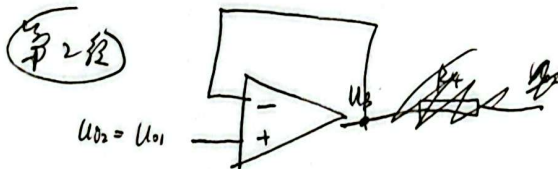
$$u_A = 0 \quad \therefore u_- = u_+ = 0$$

$$\therefore i_1 = i_A$$

$$\frac{u_{01} - u_A}{R_1} = \frac{u_A - u_{01}}{\frac{1}{sC} + R_2}, \quad Z_A = \frac{1}{\frac{1}{sC} + R_2 + R_4}$$

$$\frac{u_{01}}{R_1} = - \frac{u_{01}}{Z_A}$$

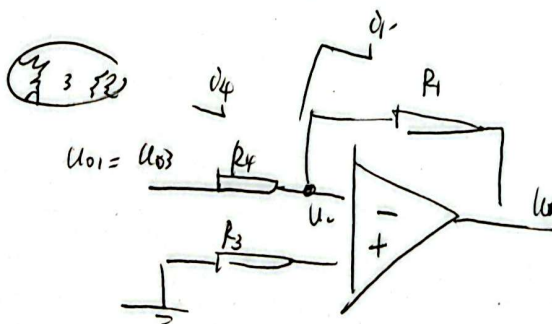
$$\frac{u_{01}}{u_{01}} = - \frac{R_1}{Z_A}$$



$$\frac{u_{01} - u_{02}}{0} = \frac{u_{03} - u_{02}}{R_4}$$

$$u_+ = u_- = u_{03} = u_{02}$$

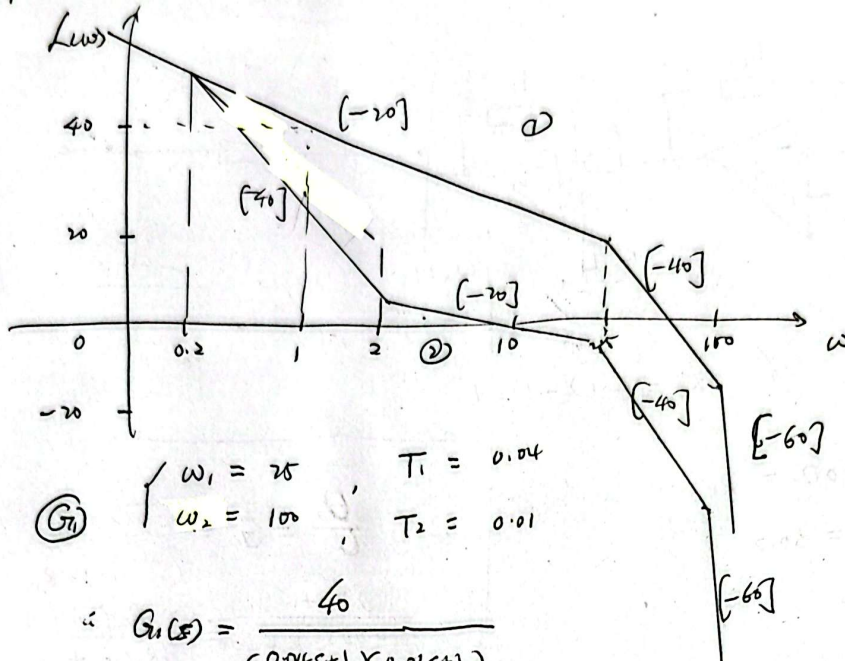
$$\therefore u_{03} = u_{02} = u_{01} = u_{01}$$



$$\frac{u_{03} - u_{01}}{R_4} = \frac{u_0 - u_{01}}{R_1}$$

$$\frac{u_0}{u_{03}} = \left[\right]$$

Q7



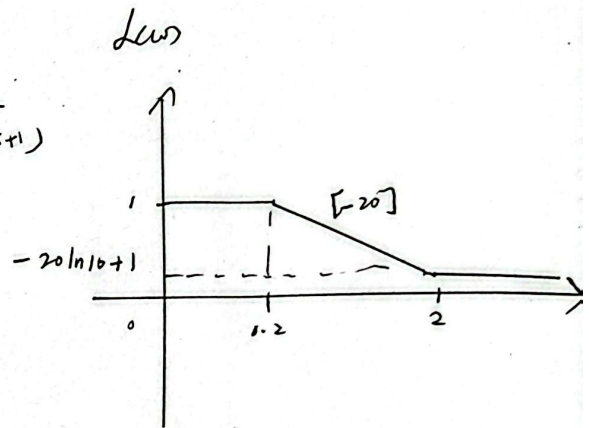
$\textcircled{G_2} \begin{cases} \omega_3 = 0.2, & T_3 = 5 \\ \omega_4 = 2, & T_4 = 0.5 \end{cases}$

$$G_2(s) = \frac{40(0.5s+1)}{(0.5s+1)(0.04s+1)(0.01s+1)}$$

$\textcircled{2} \quad G_c(s) = \frac{0.5s+1}{0.05s+1}$

$$|G_c(s)| = \frac{\sqrt{(0.5\omega)^2 + 1}}{\sqrt{(0.05\omega)^2 + 1}}$$

$$= \frac{\sqrt{0.25\omega^2 + 1}}{\sqrt{0.0025\omega^2 + 1}}$$



$$\omega \rightarrow 0, |G_c(s)| = 1$$

$$\therefore L(2) = -20 \ln\left(\frac{2}{0.2}\right) + L(0.2)$$

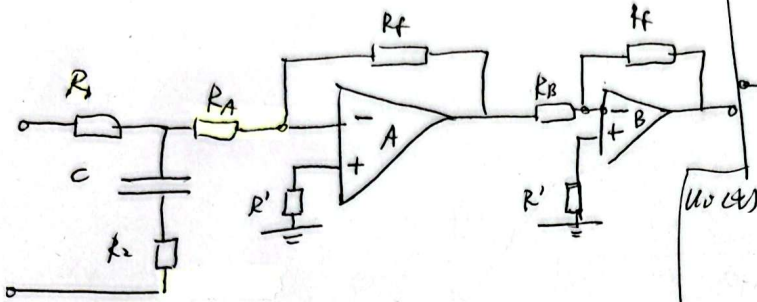
$$\therefore L(0.2) = 1$$

$$\therefore L(2) = -20 \ln 10 + 1$$

②

有源

滞后校正



$$K_A \cdot K_B = (-1)(-1) = 1$$

条件

$$R_f = R_A = 100\Omega = R_2$$

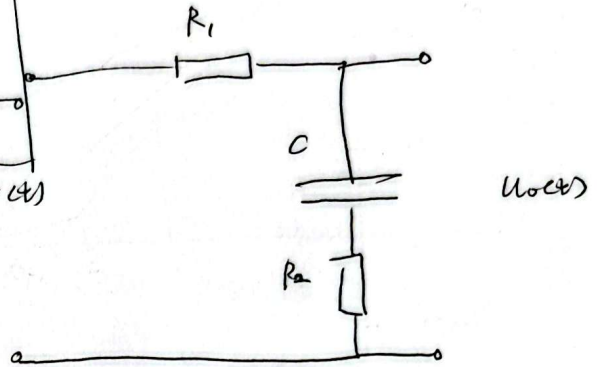
$$R' = R_f // R_A = 50\Omega$$

传递函数

$$R_2 C = 0.5$$

$$(R_1 + R_2) C = 0.05$$

无源



$$\frac{U_o}{U_i} = G_c = \frac{\frac{1}{sC} + R_2}{(\frac{1}{sC} + R_2) + R_1}$$

$$\frac{0.5s + 1}{0.05s + 1} = \frac{R_2 Cs + 1}{(R_1 + R_2)Cs + 1}$$

$$\therefore \begin{cases} R_2 C = 0.5 \\ (R_1 + R_2) C = 0.05 \end{cases}$$